

STATISTICS — COMPLETE BEGINNER'S GUIDE

Time Series Regression

From Zero to Full Understanding

Who this guide is for. This document assumes you have never studied statistics before. Every concept is built from the ground up with plain language and step-by-step worked examples.

What is new in this edition. Four improvements over the original:

- (1) **Confidence intervals** — new Section 0.4.1 explains construction and interpretation.
- (2) **Fixed ρ magnitude table** — the table in B.2 now shows what each range of ρ means in practice.
- (3) **Fixed Summary Card** — the four AR(1) cases table in C.3 now has complete Stationary? and Fix columns.
- (4) **Practice exam questions** — six fully worked exam-style questions added at the end.

NOTATION GLOSSARY — Read This First

Every symbol you will encounter, explained in plain English.

Symbol	Name	Plain Meaning
Y	Dependent variable	The thing you are trying to predict or explain
X	Independent variable	The thing you use to explain Y
t	Time index	Which time period we are in ($t = 1$ means year 1)
i	Individual index	Which person/firm/country we are looking at
β (beta)	Regression coefficient	How much Y changes when X changes by 1 unit
α (alpha)	Intercept	The value of Y when all X 's = 0
ε (epsilon)	Error / residual	The part of Y that our model fails to explain
ρ (rho)	Autocorrelation coeff.	How strongly this period's error relates to last period's
γ (gamma)	AR coeff. on lagged Y	How much last period's Y feeds into this period's Y
λ (lambda)	Trend slope	How much the mean of Y grows each period
σ^2	Variance	Measure of spread / unpredictability
Δ (delta)	First difference	Change from one period to the next: $\Delta X_t = X_t - X_{t-1}$
Σ (sigma)	Sum	Add up everything that follows
H_0	Null hypothesis	The "boring" default claim we try to disprove
H_1	Alternative hypothesis	The "interesting" claim we want to support
n	Sample size	Total number of observations
R^2	R-squared	Proportion of Y 's variation explained by the model (0 to 1)
SE	Standard error	How much a coefficient estimate varies across samples
CI	Confidence interval	A range of plausible values for the true coefficient ★ NEW
LM	Lagrange Multiplier	Test statistic to check for autocorrelation
χ^2	Chi-squared	A probability distribution used for some hypothesis tests
DW	Durbin-Watson	Quick autocorrelation check (limited use)

PART 0 — STATISTICS FOUNDATIONS

Everything in this section is background knowledge that the rest of the guide builds on.

0.1 — What Is Statistics? What Is Regression?

Statistics is the science of learning from data. We observe the world, collect numbers, and try to answer questions like: Does more education cause higher wages? Does advertising increase sales?

Regression is the central tool. It finds the relationship between explanatory variables (X's) and an outcome variable (Y).

The Simplest Regression (One Variable)

$$Y = \alpha + \beta X + \varepsilon$$

Term	Meaning	Example
Y	Outcome to explain	Hourly wage (€)
X	Thing that explains Y	Years of education
α	Baseline Y when X = 0	Wage with zero education
β	Slope — change in Y per 1-unit rise in X	€1.20 more per extra year
ε	Error — what the model misses	Luck, unobserved talent

Example: Wage = 5 + 1.2 × Education + ε . A person with 10 years of education earns an estimated 5 + 1.2 × 10 = **€17/hour**. If actual wage is €19, residual = €2.

Multiple Regression (Several Variables)

$$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \varepsilon$$

Each β_i = the *ceteris paribus* (all else equal) effect of X_i on Y.

0.2 — How Regression Is Estimated: OLS

OLS = Ordinary Least Squares. Chooses α and β to minimise the sum of squared errors across all observations:

$$\min \sum \varepsilon_i^2 = \sum (Y_i - \alpha - \beta X_i)^2$$

Why square them? (1) Positive and negative errors do not cancel. (2) Large errors are penalised more than small ones.

OLS gives coefficient estimates $\hat{\beta}_i$ ("beta hat") — the best-fitting line.

BLUE: When classical assumptions hold, OLS is the Best Linear Unbiased Estimator — no other linear method has smaller estimation errors.

0.3 — R²: How Well Does the Model Fit?

$$R^2 = 1 - (\text{Sum of Squared Errors}) / (\text{Total Variation in Y})$$

R ² Value	Meaning
0.0	Model explains nothing — X tells us nothing about Y
0.5	Model explains 50% of Y's variation
1.0	Perfect fit — model explains everything

R ² Value	Meaning
0.95	Looks great — but in time-series this can signal spurious regression ■

Intuition: Imagine Y is ocean-wave height. Total variation = how much waves differ from average. If the model explains 80%, $R^2 = 0.80$.

0.4 — Standard Errors, t-statistics, and p-values

Standard errors (SE) measure how much β varies if we collected a different sample.

$$t = \beta / SE(\beta)$$

A large $|t|$ (say > 2): estimate is far from zero — likely real.

A small $|t|$ (say < 1): estimate could easily be zero by chance.

p-value	Interpretation
$p < 0.01$	Very strong evidence against H_0 (significant at 1%)
$p < 0.05$	Strong evidence against H_0 (significant at 5%) — standard threshold
$p < 0.10$	Weak evidence against H_0 (significant at 10%)
$p > 0.10$	No convincing evidence; cannot reject H_0

Analogy: Flip a coin 20 times and get 17 heads. The p-value asks: "If fair coin (H_0), how likely is 17+ heads?" If tiny ($p < 0.05$), conclude coin is probably not fair — reject H_0 .

Important: We never "accept H_0 ". We either reject it or *fail to reject* it. Failing to reject means the data do not give us enough evidence — not that H_0 is definitely true.

★ NEW

0.4.1 — Confidence Intervals

A **confidence interval (CI)** gives a range of plausible values for the true β . The most common is the 95% CI:

$$95\% \text{ CI} = \beta \pm 1.96 \times SE(\beta)$$

What this means: If we repeated the study many times, 95% of the intervals constructed this way would contain the true β .

Three key rules:

1. If the CI does **NOT** include zero → coefficient is statistically significant at 5%.
2. If the CI **DOES** include zero → cannot reject H_0 (consistent with no effect).
3. Narrower CI = more precise estimate (comes from larger n or smaller SE).

Confidence Level	Formula	Use when
90% CI	$\beta \pm 1.645 \times SE$	Less strict threshold
95% CI	$\beta \pm 1.96 \times SE$	Standard — most common
99% CI	$\beta \pm 2.576 \times SE$	Very strict threshold

Relationship to t-test: The 95% CI excludes zero if and only if $|t| > 1.96$. They always agree — you cannot get a significant t-test but an insignificant CI, or vice versa.

Worked Example: A wage regression gives $\beta(\text{education}) = 1.20$, $SE = 0.35$, $t = 3.43$, $p = 0.001$.

95% CI = $1.20 \pm 1.96 \times 0.35 = 1.20 \pm 0.686 = [0.51, 1.89]$

Interpretation: We are 95% confident that one extra year of education raises wages by between **€0.51 and €1.89 per hour**.

Since CI excludes zero $\rightarrow$ significant at 5% ✓ (consistent with $p = 0.001$).

0.5 — The Five Classical Assumptions

OLS is BLUE (Best Linear Unbiased Estimator) only when these hold:

Assumption	Formal Statement	Why It Matters
A1	$E(\varepsilon) = 0$	Errors average to zero — no systematic over/under-prediction
A2	$\text{Var}(\varepsilon) = \sigma^2$ (constant)	Homoskedasticity — equal spread of errors everywhere
A3	$\text{cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$	Errors are uncorrelated with each other
A4	X's non-random, not perfectly collinear	We can separate each X's individual effect
A5	$\varepsilon \sim \text{Normal}$ (optional)	Needed for exact inference in small samples

Time-series warning: A3 is routinely violated in time-series data (autocorrelation). A new assumption is also needed: **stationarity**.

PART 1 — TIME SERIES FOUNDATIONS

CHAPTER TS_A — Cautions

What makes time-series data special and why ordinary regression needs modification.

A.1 — Cross-Section vs. Time-Series Data

Type	Description	Notation	Order Matters?
Cross-section	Many units at one point in time	X_i (i = individual)	No — rows interchangeable
Time-series	One unit observed repeatedly over time	X_t (t = time period)	YES — shuffling destroys information
Panel data	Many units over many time periods	X_{it}	Both dimensions matter

Why order matters: In a cross-section of 1,000 workers you could reorder rows and get the same regression result. In a time series of 40 years of GDP, row 5 must come after row 4 — the temporal structure IS the point.

A.2 — Lagged Variables

A **lag** is the value of a variable at an earlier time period — like looking in the rear-view mirror.

Notation	Meaning	Value refers to
X_t	Current value	Current period
X_{t-1}	One-period lag	One period ago
X_{t-2}	Two-period lag	Two periods ago
X_{t-q}	q -period lag	q periods ago

Visual — lag in data:

t	X_t (original)	X_{t-1} (lag 1)
1	10	— (missing, no period 0)
2	14	10 ← from $t=1$
3	11	14 ← from $t=2$
4	16	11 ← from $t=3$
5	13	16 ← from $t=4$

Why lags are needed: Economic processes rarely respond instantaneously. Safety training in month t → behaviour change → accident reduction in $t+1$, $t+2$. Central bank rate rise → affects spending over 6–12 months.

Lost observations: Lag of order q costs q observations. Starting with $n = 100$, a $DL(3)$ model leaves only 97 usable rows.

A.3 — Difference Variables

$$\Delta X_t = X_t - X_{t-1}$$

Example: GDP was €100bn last year, €104bn this year → $\Delta \text{GDP} = +€4\text{bn}$. It answers: "How much did GDP change?"

Original Variable	After Differencing	Meaning
GDP level (€bn)	ΔGDP	Change in GDP vs previous period
$\log(\text{GDP})$	$\Delta \log(\text{GDP})$	Log-return $\approx$ percentage growth rate
Unemployment (%)	$\Delta \text{unemployment}$	Change in rate (percentage points)
$\log(\text{price})$	$\Delta \log(\text{price})$	Percentage price change (return)

Why logarithms? A stock rising from €100 to €200 is +100%. Another rising from €10 to €20 is also +100%. In levels these are incomparable. $\Delta \ln(X_t) \approx$ percentage change — directly comparable across differently-scaled quantities.

Example: $\Delta \ln(100 \rightarrow 102) = \ln(1.02) \approx 2\%$ growth. ✓

Why differences matter: (1) Economic theory talks about changes. (2) They cure a stochastic trend (unit root). (3) Log-returns are scale-free.

A.4 — The Three Problems with Time-Series Regression

Problem 1 — Data quality: Time-series is data-hungry. $n > 100$ ideal; minimum ~ 35 . Use **real (deflated)** monetary values — inflation creates artificial trends that produce spurious regressions.

Problem 2 — Autocorrelation: A3 is routinely violated. Shocks persist over time. Covered in Chapter TS_B.

Problem 3 — Spurious regression: Regress Belgian ice cream sales on sunspot activity. Both trend upward over 30 years → $R^2 = 0.95$, significant at $p < 0.001$ — yet completely meaningless. Solution: ensure stationarity before fitting. Covered in Chapter TS_C.

CHAPTER TS_B — Autocorrelation

Errors that remember the past: detection with the LMSC test.

B.1 — Which Classical Assumption Is Violated?

Autocorrelation violates **A3**: $\text{cov}(\varepsilon_t, \varepsilon_{t-1}) \neq 0$.

Why A3 fails so commonly: Think of residual ε_t as "everything that moves Y but isn't in my model." Omitted variables in time series tend to persist over time (business cycles, consumer confidence, institutional inertia). Today's omitted shock carries over into tomorrow.

Intuitive example: You model consumer spending using only income. But consumer confidence (omitted) also drives spending — and confidence is persistent. Residuals form waves: positive for months (high confidence), then negative (low confidence). That is positive autocorrelation.

B.2 — First-Order Autocorrelation

The AR(1) error structure:

$$\varepsilon_t = \rho \varepsilon_{t-1} + u_t$$

where ρ (rho) is the autocorrelation coefficient ($-1 < \rho < 1$) and u_t is **white noise** — pure randomness, no memory, mean zero, constant variance. This equation says: today's error = $\rho \times$ yesterday's error + a fresh shock.

Residual patterns:

Pattern	What you see	Conclusion
No autocorrelation ($\rho = 0$)	Random scatter above and below zero — no runs	A3 satisfied ✓
Positive autocorrelation ($\rho > 0$)	Long runs of same sign: + + + - - - (waves)	A3 violated ✗
Negative autocorrelation ($\rho < 0$)	Alternating signs: + - + - + - (zigzag)	A3 violated ✗
★ FIXED		

Understanding the Size of ρ

Value of ρ	Meaning	Pattern in Residuals
$\rho = 0$	No autocorrelation. $\varepsilon_t = u_t$, purely random — A3 satisfied.	Random scatter
$0 < \rho \leq 0.3$	Mild positive autocorrelation. Small runs. Often tolerable but worth testing.	Slight clustering
$0.3 < \rho \leq 0.6$	Moderate positive autocorrelation. Clear wave pattern. LMSC very likely to detect.	Visible waves
$0.6 < \rho < 1$	Strong positive autocorrelation. Long persistent runs. Model likely misspecified.	Long same-sign runs
$\rho < 0$	Negative autocorrelation. Alternating signs. Rare; can arise from over-differencing.	Alternating $\pm \pm \pm$

Worked analogy for $\rho = 0.7$: If this quarter's residual is +10, next quarter's is expected to be +7 ($= 0.7 \times 10$). Then +4.9, +3.4, +2.4 ... slowly decaying toward zero. A shock takes many periods to fade.

We require $-1 < \rho < 1$ for the error process itself to be stationary and well-behaved.

B.3 — Consequences of Autocorrelation

Property	With Autocorrelation (static or DL models)
OLS estimates β	Still unbiased — on average they hit the true β
OLS estimates β	No longer BEST — a better estimator exists
Standard errors	WRONG — too small when $\rho > 0$ (over-confident)
t-tests and F-tests	Invalid — built on wrong standard errors
Confidence intervals	Wrong width — too narrow when $\rho > 0$ (falsely precise)
R^2	Unaffected but potentially misleading

Key danger: false significance. Positive autocorrelation inflates t-statistics, making coefficients appear more significant than they really are.

Analogy: Asking 10 people in one household whether they like pizza and claiming 10 independent data points. They're not — family members influence each other. Effective sample size is much smaller than 10. Autocorrelation does the same to time-series: consecutive observations are not truly independent.

Exception — AR(1) models: When Y_{t-1} is a regressor AND autocorrelation is present, coefficient estimates themselves become **biased** — not just inefficient. Much worse. Practical rule: if autocorrelation is detected in an AR(1) model, switch to DL(q).

B.4 — Detection: The LMSC Test

Step 0 — Graphical inspection first. Always plot residuals against time. Waves → positive AC. Zigzag → negative AC. Random scatter → A3 OK.

Hypotheses: $H_0: \rho = 0$ (no autocorrelation, A3 satisfied) vs $H_1: \rho \neq 0$ (autocorrelation present, A3 violated)

Step 1: Fit the original model. Save residuals e_t .

Step 2: Auxiliary regression — regress residuals on all original predictors PLUS the lagged residual:

$$e_t = \alpha_0 + \alpha_1 X_{1t} + \dots + \alpha_k X_{kt} + \alpha_{k+1} e_{t-1} + u_t$$

Why include the original X's? Without them, some of the relationship between e_t and e_{t-1} might be caused by the X variables themselves (which also have temporal structure). Including X's "controls out" their influence, leaving only the pure autocorrelation signal.

Step 3: $LM = n \times R^2$ where $n = n_{\text{original}} - 1$

Step 4: Under H_0 , $LM \sim \chi^2(1)$. Compare to **3.841** (the 5% critical value for $\chi^2(1)$).

Decision: $LM > 3.841 \rightarrow$ Reject $H_0 \rightarrow$ autocorrelation present.

$LM \leq 3.841 \rightarrow$ Do not reject $H_0 \rightarrow$ A3 plausibly satisfied.

Worked Example — Phillips Curve (n = 90 quarterly):

Item	Value
n for auxiliary regression	$90 - 1 = 89$
R^2 of auxiliary regression	0.310211
Coefficient on e_{t-1}	0.558 (t = 6.219, p < 0.001)
LM statistic	$89 \times 0.310211 = 27.609$

Item	Value
Critical value (5%)	3.841
Conclusion	27.609 >> 3.841 → Reject H_0 . Strong autocorrelation.

Durbin-Watson as a quick alternative:

DW Value	Interpretation
Close to 2	No autocorrelation
Close to 0	Strong positive autocorrelation (ρ near +1)
Close to 4	Strong negative autocorrelation (ρ near -1)
Between 1.5 and 2.5	Generally acceptable (rough rule)

$DW \approx 2(1 - \rho)$. **Critical limitation:** DW is invalid whenever Y_{t-1} appears as a regressor. Always use LMSC as the primary test.

B.5 — Solutions for Autocorrelation

The philosophy: Autocorrelation in a static model usually indicates model misspecification — the error is carrying memory because dynamics were left out. Adding lags absorbs the dynamic structure.

Re-specification	What you add	Model type
Add lagged X's	$X_{t-1}, X_{t-2}, \dots, X_{t-q}$	DL(q) — Distributed Lag
Add lagged Y	Y_{t-1}	AR(1) — Autoregressive

DL(q) vs AR(1): DL is preferred when autocorrelation is present — it remains unbiased (only inefficient). AR(1) is preferred for parsimony when A3 is already satisfied. **Key principle: bias is worse than inefficiency.**

CHAPTER TS_C — Stationarity

Spurious regression, the generalised AR(1), and the Dickey-Fuller test.

C.1 — Spurious Regression

The problem: Regress Belgian GDP per capita (1950–2004) on US population density. Result: $R^2 \approx 0.96$, $p < 0.001$ — yet no meaningful economic relationship exists. This is a **spurious regression**.

Famous example: Nicolas Cage films released per year vs US swimming pool drownings (1999–2009): $R^2 \approx 0.67$, $p < 0.05$. Pure coincidence. Both series happened to move similarly over that period.

Root cause: Both variables are non-stationary — their means drift over time. Standard regression theory requires stationarity. When it fails, t-statistics don't follow t-distributions and R^2 is meaningless.

Stationarity is therefore a **required additional assumption** for time-series regression.

C.2 — Definition of Stationarity

A time series Y_t is **weakly stationary** if for every t :

1. $E(Y_t) = \mu$ — **constant mean**
2. $\text{Var}(Y_t) = \sigma^2$ — **constant variance**
3. $\text{Cov}(Y_t, Y_{t-s}) = \gamma_s$ — **covariance depends only on lag s**

Series Type	Visual Pattern	Stationary?
Stationary	Fluctuates around a fixed level, similar spread throughout	YES ✓
Non-stationary (trend)	Drifts steadily upward (or downward) over time	NO ✗
Non-stationary (random walk)	Wanders unpredictably — goes wherever past shocks pushed it	NO ✗

"Cut in half" test: Divide the series into two halves. If both halves have similar mean, similar spread, and similar patterns → likely stationary. If the right half is systematically higher, more volatile, or different in pattern → evidence of non-stationarity.

C.3 — The Generalised AR(1) Model

$$Y_t = a + \lambda t + \rho Y_{t-1} + \varepsilon_t$$

Three parameters determine everything: a (constant), λ (trend slope), ρ (autoregressive coefficient). The critical question: is $|\rho| < 1$ (stationary family) or $\rho = 1$ (random walk)?

The Four Cases

Case	Model	Properties	Stationary?
Case 1 — Basic AR(1) $ \rho < 1$, $a=0$, $\lambda=0$	$Y_t = \rho Y_{t-1} + \varepsilon_t$	Mean = 0. Var = $\sigma^2/(1-\rho^2)$. Shocks decay geometrically. Series returns to 0.	YES ✓
Case 2 — AR(1) + constant $ \rho < 1$, $a \neq 0$, $\lambda=0$	$Y_t = a + \rho Y_{t-1} + \varepsilon_t$	Long-run mean $\mu = a/(1-\rho)$. Series oscillates around μ .	YES ✓
Case 3 — Deterministic trend $ \rho < 1$, $a \neq 0$, $\lambda \neq 0$	$Y_t = a + \lambda t + \rho Y_{t-1} + \varepsilon_t$	Mean grows linearly. Deviations from trend line ARE stationary. Fix: add t as regressor.	NO ✗ (in levels)

Case	Model	Properties	Stationary?
Case 4 — Random walk $\rho = 1$	$Y_t = a + Y_{t-1} + \varepsilon_t$	Each shock is PERMANENT. $\text{Var}(Y_t) = t\sigma^2 \rightarrow$ grows without bound. Fix: first differences.	NO ✗

★ FIXED

Summary Card — The Four Cases at a Glance

Case	ρ	λ	Stationary?	Fix
Basic AR(1)	$ \rho < 1$	0	YES ✓	None needed — use in levels
AR(1) + constant	$ \rho < 1$	0	YES ✓	None needed — use in levels
AR(1) + deterministic trend	$ \rho < 1$	$\neq 0$	NO ✗ (in levels)	Add t as regressor in main model
Random walk	$\rho = 1$	any	NO ✗	Take first differences (ΔY_t)

C.4 — Deterministic vs. Stochastic Trends

This distinction is crucial — the two types of non-stationarity require **different fixes**. Applying the wrong fix can make things worse.

Feature	Deterministic Trend	Stochastic Trend (Random Walk)
Mathematical source	λt term in the model	$\rho = 1$
Mean	Changes predictably (grows by λ per period)	May drift, but unpredictably
Variance	Constant	Grows with time
Shocks	Temporary — returns to trend after shock	Permanent — series never recovers
Example	GDP trend growth of 2% per year	S&P; 500 stock price
Correct fix	Add t as regressor (or detrend)	First difference
Wrong fix	First-differencing $\rightarrow$ over-differences $\rightarrow$ negative AC	Detrending does not remove stochastic trend

Key intuition — permanence of shocks:

Deterministic: A recession pushes GDP below its trend, but in subsequent years GDP returns to the same trend. Shock is temporary.

Random walk: A recession permanently lowers the level. All future values are shifted down. Shock is permanent and the series has no "memory" of where it "should" be.

C.5 — Solutions for Non-Stationarity

Fix A — Deterministic trend ($|\rho| < 1, \lambda \neq 0$): Include t as an additional regressor:

$$Y_t = \beta_0 + \beta_1 X_t + \lambda t + \varepsilon_t$$

The coefficient on t absorbs the linear trend. The coefficient on X_t then measures the relationship *after controlling for the common time trend*. Seasonal: use $\Delta_{12} X_t = X_t - X_{t-12}$ to remove annual seasonality.

Fix B — Stochastic trend / unit root ($\rho = 1$): Take first differences:

$$\Delta Y_t = Y_t - Y_{t-1} \text{ and } \Delta X_t = X_t - X_{t-1}$$

If $Y_t = Y_{t-1} + \varepsilon_t$, then $\Delta Y_t = \varepsilon_t$ — white noise, stationary ✓. The difference operation strips away accumulated shock history.

Cointegration caveat: If two non-stationary series are genuinely tied together economically, do NOT difference them. Run the regression in levels (the "cointegrating regression"). Differencing cointegrated variables throws away the most economically meaningful information.

C.6 — The Dickey-Fuller Test

Purpose: Formal unit root test — decides whether $\rho = 1$ (non-stationary) or $|\rho| < 1$ (stationary).

Hypotheses: $H_0: \rho = 1$ (unit root — NOT stationary) vs $H_A: |\rho| < 1$ (stationary)

Reparameterisation: Subtract Y_{t-1} from both sides $\rightarrow \alpha = \rho - 1$. $H_0: \alpha = 0$. $H_A: \alpha < 0$ (one-sided test).

$$\Delta Y_t = a + \lambda t + \alpha Y_{t-1} + \varepsilon_t$$

Why the normal t-table cannot be used: Under H_0 , Y_t is a random walk — non-stationary, variance grows over time. Standard asymptotic theory breaks down. The DF t-distribution is shifted LEFT compared to the usual t-distribution. Using ± 1.96 would almost never reject H_0 even for clearly stationary series.

DF Version	Regression	1% CV	5% CV	10% CV	Use when
V1 — no trend	$\Delta Y_t = \alpha + \alpha Y_{t-1} + \varepsilon_t$	-3.43	-2.86	-2.57	No visible trend in time plot
V2 — with trend	$\Delta Y_t = \alpha + \lambda t + \alpha Y_{t-1} + \varepsilon_t$	-3.96	-3.41	-3.13	Clear trend or unsure

Decision rule: Reject H_0 if the t-statistic on Y_{t-1} is MORE NEGATIVE than the critical value.

Example: $t = -3.10$. V1 at 5%: $-3.10 < -2.86 \rightarrow$ Reject $H_0 \rightarrow$ stationary ✓. V2 at 5%: $-3.10 > -3.41 \rightarrow$ Fail to reject $\rightarrow$ unit root ✗. The choice of version matters!

V2 is the safer default: If a trend is present but you use V1, the test is misspecified. If no trend is present but you use V2, you merely lose one degree of freedom.

CHAPTER TS_D — Dynamic Models

DL(q), AR(1), total multipliers, and the time-series exam workflow.

D.1 — Distributed Lag Model DL(q)

A static model assumes the entire effect of X on Y is instantaneous. But most economic effects are delayed. The DL(q) model captures this:

$$Y_t = \alpha + \beta_0 X_t + \beta_1 X_{t-1} + \beta_2 X_{t-2} + \dots + \beta_q X_{t-q} + \varepsilon_t$$

The effect of X is **distributed across q+1 periods** — hence the name.

Type of Change	Effect at Lag k	Long-Run Total
Temporary (X rises by 1, then returns)	β_k is the additional effect k periods later	Effect ends after lag q
Maintained (X rises by 1 permanently)	Cumulative: $\beta_0 + \beta_1 + \dots + \beta_k$ after k periods	Total multiplier = $\sum \beta_k$

$$\text{Total Multiplier} = \beta_0 + \beta_1 + \dots + \beta_q$$

Interpretation: The total multiplier is the long-run effect on Y of a permanent 1-unit increase in X. It answers: "If X goes up by 1 and stays there forever, by how much will Y eventually change?"

D.2 — Choosing the Optimal Lag Length q

The top-down procedure: Start from $q_{\max}$, progressively drop the highest lag that is not significant.

1. Fit DL($q_{\max}$). Test $\beta_{q_{\max}}$.
2. If not significant: drop lag $q_{\max}$, refit DL($q_{\max} - 1$).
3. Repeat until the highest remaining lag IS significant.
4. That is your optimal q.

Important: Never drop intermediate insignificant lags. If β_k is significant but β_{k+1} is not, keep both — you cannot have lag 2 without lag 1.

Worked Example — Safety Training ($q_{\max} = 4$):

Step	Model	Result	Action
1	DL(4)	$\beta_4: t = 0.83, p = 0.41$	Not significant → drop lag 4
2	DL(3)	$\beta_3: t = 1.21, p = 0.23$	Not significant → drop lag 3
3	DL(2)	$\beta_2: t = 1.56, p = 0.12$	Not significant → drop lag 2
4	DL(1)	$\beta_1: t = -2.87, p = 0.005$	SIGNIFICANT → Stop. Optimal q = 1

Final model: $Y_t = \alpha + \beta_0 X_t + \beta_1 X_{t-1}$ | Total multiplier = $\beta_0 + \beta_1$

D.3 — Autoregressive Model AR(1)

$$Y_t = \alpha + \beta X_t + \gamma Y_{t-1} + \varepsilon_t$$

Instead of past X's, this model uses **past Y** to carry the memory of the system — capturing all persistence in a single coefficient γ .

Coefficient	Interpretation
α	Intercept (baseline)

Coefficient	Interpretation
β	Immediate effect: a 1-unit rise in X_t raises Y_t by β , holding Y_{t-1} fixed
γ	Persistence: how much of last period's Y carries forward. Requires $ \gamma < 1$.
$\beta/(1-\gamma)$	Total multiplier: long-run effect of a permanent 1-unit rise in X

★ NEW — ENHANCED

Why $|\gamma| < 1$ Is Required — Full Explanation

This condition appears in three related contexts. Here is what it means in each:

1. In the pure time-series AR(1) (Chapter C.3):

If $|\rho| \geq 1$, each shock to Y is never absorbed — it persists forever or grows. The variance of $Y_t = \tau\sigma^2$ (grows without bound). The series wanders without returning to a fixed level. This is the definition of non-stationarity.

2. In the AR(1) regression model:

When you write $Y_t = \alpha + \beta X_t + \gamma Y_{t-1} + \varepsilon_t$, the condition $|\gamma| < 1$ plays the same role. Each period, Y partially reverts to its long-run mean. The formula $\mu = \alpha/(1-\gamma)$ only makes sense when $|\gamma| < 1$ — otherwise you divide by zero or a negative number, producing a nonsensical mean.

3. For the total multiplier $\beta/(1-\gamma)$:

This formula comes from summing the geometric series $\beta + \beta\gamma + \beta\gamma^2 + \beta\gamma^3 + \dots$. If $|\gamma| \geq 1$, the terms do not shrink — the sum is infinite. A permanent 1-unit increase in X would have an infinite long-run effect, which is economically impossible. The condition $|\gamma| < 1$ guarantees the sum converges to the finite value $\beta/(1-\gamma)$.

γ value	Implication	Action
$ \gamma < 1$	Stationary. Total multiplier = $\beta/(1-\gamma)$. Safe to interpret.	✓ Proceed normally
$\gamma = 1$	Explosive / unit root in Y . AR(1) inappropriate.	Take first differences instead
$ \gamma > 1$	Series would explode over time. Severe misspecification.	Re-specify the model

Practical check: Always verify $|\gamma| < 1$ after fitting an AR(1). If $|\gamma| \geq 1$, something has gone wrong — usually non-stationary input variables or model misspecification.

Total multiplier proof: In long-run equilibrium $Y^* = Y_{t-1} = Y_t$:
 $Y^* = \alpha + \beta X^* + \gamma Y^* \rightarrow Y^*(1-\gamma) = \alpha + \beta X^* \rightarrow Y^* = \alpha/(1-\gamma) + [\beta/(1-\gamma)]X^*$
So a permanent 1-unit rise in X^* raises Y^* by $\beta/(1-\gamma)$.

Worked Example — Education Spending and GDP Growth:

Fitted: $\hat{Y}_t = 1.01 + 0.009X_t + 0.627Y_{t-1}$

Component	Value	Meaning
α	1.01	Baseline GDP growth when $X = 0$ and $Y_{t-1} = 0$
β	0.009	\$1 rise in spending per child $\rightarrow$ +0.009pp GDP growth immediately
γ	0.627	62.7% of last year's GDP growth persists into this year
Total multiplier	$0.009/(1-0.627) = 0.024$	Permanent \$1/child rise $\rightarrow$ +0.024pp long-run GDP growth

D.4 — Link Between AR(1) and DL(∞)

By repeated substitution into the AR(1) equation, we can show it equals a $DL(\infty)$ with geometrically declining coefficients:

$$Y_t = \alpha/(1-\gamma) + \beta X_t + \beta\gamma X_{t-1} + \beta\gamma^2 X_{t-2} + \beta\gamma^3 X_{t-3} + \dots$$

Total multiplier from geometric series: $\sum \beta\gamma^k = \beta \times 1/(1-\gamma) = \beta/(1-\gamma)$ (requires $|\gamma| < 1$)

Feature	DL(q)	AR(1)
Parameters needed	$q+1$	2
Multicollinearity	High for large q	Low (only one lag of Y)
Bias when A3 violated	None — only inefficient	Biased $\times$
Lag flexibility	Completely free	Constrained to geometric decay
When to prefer	Autocorrelation suspected	A3 satisfied, parsimony needed

D.5 — The Three-Step Exam Workflow

Step 1 — Stationarity Check (TS_C): For each variable:

1. Plot series against time. Does it trend? Does variance change?
2. Choose DF version: no trend visible $\rightarrow$ V1. Clear trend or unsure $\rightarrow$ V2.
3. Run DF regression, read t-stat on Y_{t-1} .
4. Compare to adapted critical values (-2.86 V1, -3.41 V2 at 5%). NOT ± 1.96 .
5. Reject $H_0 \rightarrow$ stationary $\rightarrow$ use in levels.
6. Fail to reject $\rightarrow$ unit root $\rightarrow$ take ΔY_t , re-run DF on differences.

Step 2 — Fit the Model (TS_D):

For $DL(q)$: start at $q_{\max}$, drop highest non-significant lag until highest is significant. Report all coefficients. Total multiplier = $\sum \beta\gamma^k$.

For $AR(1)$: fit directly. Report β , γ , and total multiplier = $\beta/(1-\gamma)$. Verify $|\gamma| < 1$.

Step 3 — LMSC Assumption Check (TS_B):

1. State $H_0: \rho = 0$ and $H_A: \rho \neq 0$.
2. Write auxiliary regression (all X's + lagged residual e_{t-1}).
3. Compute $LM = n_{\text{aux}} \times R^2_{\text{aux}}$.
4. Compare to $\chi^2(1) = 3.841$.
5. Write a full conclusion sentence.

Worked Example — Consumption and Income ($n = 162$ quarterly):

Both $\ln(Y)$ and $\ln(X)$ show clear upward trends.

Variable	DF Version	t-statistic	Critical (5%)	Conclusion
$\ln(Y)$ — level	V2 (trend)	-1.82	-3.41	Fail to reject $\rightarrow$ unit root
$\ln(X)$ — level	V2 (trend)	-1.65	-3.41	Fail to reject $\rightarrow$ unit root
$\Delta \ln(Y)$ — diff	V1 (no trend)	-9.95	-2.86	Reject $\rightarrow$ stationary $\checkmark$
$\Delta \ln(X)$ — diff	V1 (no trend)	-12.72	-2.86	Reject $\rightarrow$ stationary $\checkmark$

Top-down $DL(q)$, $q_{\max} = 3$:

Model	Highest Lag t-stat	Significant?	Action
DL(3)	$\beta_3: t = 0.91$	No	Drop lag 3
DL(2)	$\beta_2: t = 1.38$	No	Drop lag 2
DL(1)	$\beta_1: t = 2.74$	YES	Stop. Optimal $q = 1$

Fitted: $\Delta X_t = 0.004 + 0.350\Delta X_t + 0.182\Delta X_{t-1}$ | Total multiplier = 0.532

LMSC: $n_{aux} = 161$, $R^2 = 0.002618$, $LM = 0.4215 < 3.841 \rightarrow$ Fail to reject H_0 . A3 plausibly satisfied. ✓

FORMULA SHEET — The Five Equations You Must Know

1. DL(q) Model

$$Y_t = \alpha + \beta_1 X_t + \beta_2 X_{t-1} + \dots + \beta_q X_{t-q} + \varepsilon_t \quad \text{Total multiplier} = \beta_1 + \beta_2 + \dots + \beta_q = \sum \beta_i$$

2. AR(1) Model

$$Y_t = \alpha + \beta X_t + \gamma Y_{t-1} + \varepsilon_t \quad \text{Total multiplier} = \beta/(1-\gamma) \quad [\text{requires } |\gamma| < 1]$$

3. LMSC Test

$$\text{Auxiliary: } e_t = \alpha_0 + \alpha_1 X_{1t} + \dots + \alpha_k X_{kt} + \alpha_{k+1} e_{t-1} + u_t \quad LM = n \times R^2 \sim \chi^2(1) \\ \text{under } H_0: \rho = 0 \mid H_1: \rho \neq 0 \mid \text{Critical value at 5\%: 3.841}$$

4. Dickey-Fuller Test

$$\text{V1 (no trend): } \Delta Y_t = \alpha_0 + \alpha_1 Y_{t-1} + \varepsilon_t \quad \text{V2 (with trend): } \Delta Y_t = \alpha_0 + \lambda t + \alpha_1 Y_{t-1} + \varepsilon_t \\ H_0: \alpha_1 = 0 \text{ (unit root)} \mid H_1: \alpha_1 < 0 \text{ (stationary)} \quad \text{Critical values at 5\%: V1} = -2.86 \quad \text{V2} = -3.41$$

5. Confidence Interval ★ NEW

$$95\% \text{ CI} = \beta_1 \pm 1.96 \times SE(\beta_1) \quad 99\% \text{ CI} = \beta_1 \pm 2.576 \times SE(\beta_1) \quad \text{CI excludes zero} \leftrightarrow |t| > 1.96 \leftrightarrow p < 0.05 \text{ (all equivalent at 5\%)}$$

EXAM-DAY CHECKLIST

Step 1 — Stationarity (TS_C)

- Plot every variable against time
- Choose DF version (V1: no trend / V2: trend or unsure)
- Run DF on each variable, read t-stat on Y_{t-1}
- Compare to adapted critical values — NOT ± 1.96 or ± 2.58
- Unit root present? → take ΔY_t , re-test on differences

Step 2 — Fit the Model (TS_D)

- Model type given in question (static / DL(q) / AR(1))
- For DL: start at $q_{\max}$, drop top lag while not significant
- Report all coefficients with SEs and t-stats
- Calculate total multiplier ($\sum \beta$ or $\beta/(1-\gamma)$)
- Verify $|\gamma| < 1$ for AR(1) models
- Interpret: temporary vs. maintained change; levels vs. log-returns

Step 3 — Assumption Check (TS_B)

- State H_0 and H_A explicitly
- Write the auxiliary regression (all original X's + lagged residual)
- Compute $LM = n_{\text{aux}} \times R^2_{\text{aux}}$
- Compare to $\chi^2(1) = 3.841$ (or use p-value)
- Write a full conclusion sentence

Extra — Confidence Intervals ★ NEW

- $CI = \beta \pm 1.96 \times SE$
- Check whether CI excludes zero (= significant at 5%)
- Interpret in context: "We are 95% confident that ..."

COMMON EXAM PITFALLS

1. "Reject H_0 means non-stationary" — WRONG

For the Dickey-Fuller test: Reject H_0 means the series IS stationary (H_0 is the unit root). For the LMSC test: Reject H_0 means autocorrelation IS present (H_0 is no autocorrelation). These are opposite! Know which test you are running.

2. "Take differences whenever the series trends" — BE CAREFUL

Only difference if the DF test confirms a stochastic trend (unit root, $\rho = 1$). If the trend is deterministic ($|\rho| < 1$, $\lambda \neq 0$), add t as a regressor instead. Over-differencing creates artificial negative autocorrelation and loses information.

3. "Use Durbin-Watson for all models" — WRONG

DW is invalid whenever Y_{t-1} appears as a regressor (AR(1) models). Always use LMSC — it works for static, DL(q), and AR(1) models.

4. "Total multiplier = β_0 only" — WRONG

For DL(q): total multiplier = $\beta_0 + \beta_1 + \dots + \beta_q$ (sum of ALL lag coefficients). For AR(1): total multiplier = $\beta_0/(1-\gamma)$, not just β_0 . β_0 alone is the immediate effect only.

5. "Unit changes when X is log-differenced" — WRONG

If you differenced $\log(X)$, a 1-unit rise in $\Delta \ln(X)$ means a 1 percentage point rise in the growth rate — not a 1-unit rise in the level. Always interpret in terms of percentage changes.

6. "Skip the autocorrelation test to save time" — WRONG

The LMSC test is worth dedicated exam marks. Always report all five elements: H_0/H_1 , the auxiliary regression equation, $LM = n \times R^2$, comparison to 3.841, and a conclusion sentence.

7. "Use the regular t-table for the Dickey-Fuller t-statistic" — WRONG

The DF t-statistic does NOT follow a t-distribution under H_0 . Standard critical values (± 1.96 , ± 2.58) do not apply. Always use adapted DF critical values: -2.86 (V1, 5%), -3.41 (V2, 5%).

8. "Confidence interval includes zero = we accept H_0 " — WRONG ★ NEW

We never accept H_0 . If the CI includes zero, we fail to reject H_0 — meaning the data are insufficient to confirm an effect, not that there is definitely no effect. If the CI excludes zero, we reject H_0 and conclude a significant effect.

PART 2 — PRACTICE EXAM QUESTIONS ★ NEW

Six fully worked exam-style questions. Attempt each question before reading the solution. Cover the answer and work through it independently — this is essential for building exam confidence.

Question 1 — Hypothesis Testing and Confidence Intervals

A researcher regresses monthly restaurant sales (€000s) on advertising spend (€000s) using $n = 48$ monthly observations. Regression output:

Coefficient	Estimate	Standard Error
Intercept (α)	42.3	8.1
β (advertising)	6.8	2.1
R^2	0.41	—

- (a) Calculate the t-statistic for the advertising coefficient.
- (b) Is advertising statistically significant at 5%? At 1%? Justify.
- (c) Construct a 95% confidence interval for β (advertising).
- (d) Interpret the confidence interval in plain English.
- (e) What does $R^2 = 0.41$ tell us?

SOLUTION

(a) $t = \beta / SE = 6.8 / 2.1 = 3.24$

(b) At 5%: $|t| = 3.24 > 1.96 \rightarrow$ significant ✓

At 1%: $|t| = 3.24 > 2.576 \rightarrow$ also significant at 1% ✓

(c) 95% CI = $6.8 \pm 1.96 \times 2.1 = 6.8 \pm 4.12 = [2.68, 10.92]$

(d) We are 95% confident that each additional €1,000 of advertising spend raises monthly sales by between **€2,680 and €10,920** on average. Since the entire CI is above zero, the effect is statistically significant.

(e) Advertising spend explains **41%** of the variation in monthly sales. The remaining 59% is explained by other factors not in this model.

Question 2 — Dickey-Fuller Test

You have quarterly Belgian unemployment data ($n = 160$). The series shows a clear upward trend. You run the Dickey-Fuller Version 2 regression and obtain: $\alpha = -0.063$, $SE(\alpha) = 0.028$. You then test the first-differenced series: t-statistic on $\Delta U_{\{t-1\}} = -11.42$.

- (a) State H_0 and H_1 for the Dickey-Fuller test.
- (b) Calculate the t-statistic for α in the level regression.
- (c) Why is Version 2 appropriate here?
- (d) Can you reject H_0 at 5%? At 1%?
- (e) What transformation should be applied? Why?
- (f) Is the transformed series stationary?

SOLUTION

(a) $H_0: \alpha = 0$ (unit root — NOT stationary)
 $H_1: \alpha < 0$ (stationary)

(b) $t = -0.063 / 0.028 = -2.25$

(c) The series shows a clear upward trend in the time plot. Version 2 includes t as a regressor, controlling for the deterministic trend component. Using Version 1 when a trend is present would misspecify the test.

(d) At 5%: critical value = -3.41 . $t = -2.25 > -3.41 \rightarrow$ **Fail to reject H_0** . Not significant at 5%.
 At 1%: critical value = -3.96 . Also fail to reject.

(e) Unemployment has a stochastic trend (unit root). Apply **first differences**: $\Delta U_t = U_t - U_{t-1}$. This converts levels into changes in the unemployment rate (percentage points), which is stationary.

(f) DF on ΔU_t : $t = -11.42$. Using Version 1 (differences show no trend): critical value = -2.86 . Since $-11.42 < -2.86 \rightarrow$ **Reject H_0** $\rightarrow \Delta U_t$ is **stationary** ✓. Use first differences in the model.

Question 3 — LMSC Autocorrelation Test

A researcher fits a DL(2) model on $n = 80$ monthly observations. To test for autocorrelation, they run an auxiliary regression and obtain $R^2 = 0.143$.

- (a) Write out the auxiliary regression equation for the LMSC test.
- (b) What is the sample size of the auxiliary regression? Why?
- (c) Compute the LM test statistic.
- (d) State H_0 and H_1 .
- (e) What is the critical value at 5%?
- (f) State your conclusion. What should the researcher do next?

SOLUTION

(a) $e_t = \alpha + \alpha_1 X_t + \alpha_2 X_{t-1} + \alpha_3 X_{t-2} + \alpha_4 e_{t-1} + u_t$

(b) $n_{\text{aux}} = 80 - 1 = 79$. We lose one observation because e_{t-1} is undefined for the first residual (no "period zero" residual).

(c) $LM = n_{\text{aux}} \times R^2 = 79 \times 0.143 = 11.30$

(d) $H_0: \rho = 0$ (no autocorrelation — A3 satisfied)
 $H_1: \rho \neq 0$ (autocorrelation present — A3 violated)

(e) $\chi^2(1, 5\%) = 3.841$. Under H_0 , $LM \sim \chi^2(1)$.

(f) $LM = 11.30 > 3.841 \rightarrow$ **Reject H_0 at 5%**. Autocorrelation is significantly present; A3 is violated. The researcher should re-specify the model by adding more lagged X's (extend to higher DL order). DL is preferred over AR(1) when autocorrelation is detected, as AR(1) would produce biased estimates.

Question 4 — DL(q) Lag Selection and Total Multiplier

A researcher models monthly workplace accidents (Y) as a function of safety training hours (X), with $q_{\text{max}} = 3$. Top-down results:

Model	Coefficient	t-statistic	p-value
DL(3)	β	0.71	0.48
DL(2)	β	1.44	0.15

Model	Coefficient	t-statistic	p-value
DL(1)	β_1	-3.12	0.003

The final model gives: $\hat{X}_t = 52.3 - 1.8X_t - 1.2X_{t-1}$

- What is the optimal lag length q ? Explain the top-down procedure.
- Write out the final chosen model equation.
- Interpret $\beta_1 = -1.8$ and $\beta_2 = -1.2$.
- Calculate and interpret the total multiplier.
- The company permanently increases training by 2 hours/month. What is the predicted long-run change in monthly accidents?

SOLUTION

- Optimal $q = 1$. Top-down: Start at DL(3). $\beta_1: t = 0.71$, not significant $\rightarrow$ drop lag 3. Refit DL(2). $\beta_1: t = 1.44$, not significant $\rightarrow$ drop lag 2. Refit DL(1). $\beta_1: t = -3.12$, significant ($|t| > 2$) $\rightarrow$ STOP.
- $\hat{X}_t = 52.3 - 1.8X_t - 1.2X_{t-1}$
- $\beta_1 = -1.8$: One additional training hour this month is associated with **1.8 fewer accidents this month**, holding last month's training constant.
 $\beta_2 = -1.2$: One additional training hour last month is associated with **1.2 fewer accidents this month**, holding this month's training constant.
- Total multiplier $= -1.8 + (-1.2) = -3.0$. A permanent 1-hour/month increase in training is associated with **3.0 fewer accidents per month** in the long run.
- $\Delta X = +2$ permanently. Long-run change $= 2 \times (-3.0) = -6$ **accidents/month** predicted reduction.

Question 5 — AR(1) Model Full Interpretation

A researcher models quarterly consumer spending (€bn) as a function of disposable income (€bn):

$$\hat{Y}_t = 12.4 + 0.31X_t + 0.74Y_{t-1}$$

- Interpret each coefficient (α , β , γ).
- Is the stationarity condition satisfied? Why does this matter?
- Calculate the total multiplier.
- Income permanently increases by €5bn. What is the predicted long-run change in spending?
- Instead, income temporarily rises by €5bn only in quarter t . Trace the additional effect for periods $t, t+1, t+2, t+3$.
- The LMSC test gives $LM = 8.4$. What is your conclusion?

SOLUTION

- $\alpha = 12.4$: baseline spending level.
 $\beta = 0.31$: A €1bn rise in income this quarter raises spending by **€0.31bn immediately**, holding last quarter's spending fixed.
 $\gamma = 0.74$: **74% of last quarter's spending level persists** into this quarter — capturing habits, contracts, and inertia.
- $|\gamma| = 0.74 < 1 \rightarrow$ stationarity condition **satisfied** ✓.
This matters because: (1) the long-run mean is finite; (2) the total multiplier formula converges; (3) the model is economically interpretable. If $\gamma \geq 1$, the model implies explosive, ever-growing spending — impossible.
- Total multiplier $= 0.31 / (1 - 0.74) = 0.31 / 0.26 = 1.19$

(d) Long-run $\Delta Y = 5 \times 1.19 = \text{€}5.96\text{bn}$ increase in spending.

(e) Temporary shock $\Delta X = +5$ in period t only:

Period	Additional Effect on Y
t	$5 \times 0.31 = +1.55$
$t+1$	$0.74 \times 1.55 = +1.15$
$t+2$	$0.74^2 \times 1.55 = +0.85$
$t+3$	$0.74^3 \times 1.55 = +0.63$
$t+k$	$0.74^k \times 1.55 \rightarrow 0$ (geometric decay)

(f) $LM = 8.4 > 3.841 \rightarrow$ **Reject H_0** . Autocorrelation significantly present, A3 violated. This is serious for AR(1): when autocorrelation is present, coefficient estimates become **biased**. The researcher should switch to a **DL(q) model**, which remains unbiased even with autocorrelation.

Question 6 — Full Three-Step Workflow

A researcher studies Belgian real GDP growth (Y) and real government investment (X) using quarterly data ($n = 104$). Both series are in log form and show upward trends.

Step 1 — DF results:

Variable	DF Version	t-statistic
$\ln(Y)$ — level	V2	-2.01
$\Delta \ln(Y)$ — diff	V1	-8.73
$\ln(X)$ — level	V2	-1.78
$\Delta \ln(X)$ — diff	V1	-9.12

Step 2 — Model fitting ($q_{\max} = 2$):

DL(2): $\beta_{\text{level}} t = 0.88$ ($p = 0.38$) $\rightarrow$ DL(1): $\beta_{\text{level}} t = 2.51$ ($p = 0.014$)

Final: $\Delta \ln(Y)_t = 0.003 + 0.28 \Delta \ln(X)_t + 0.19 \Delta \ln(X)_{t-1}$

Step 3 — LMSC: Auxiliary $R^2 = 0.018$, $n_{\text{aux}} = 101$

(a) Are $\ln(Y)$ and $\ln(X)$ stationary in levels? Show full reasoning.

(b) Are the differenced series stationary?

(c) Why is Version 1 used for the differenced series?

(d) Why does top-down give optimal $q = 1$?

(e) Interpret $\beta_{\text{level}} = 0.28$ and $\beta_{\text{level}} = 0.19$ in economic terms.

(f) Calculate and interpret the total multiplier.

(g) Conduct the full LMSC test. Is A3 satisfied?

SOLUTION

(a) $\ln(Y)$: $t = -2.01 > -3.41$ (V2, 5%) $\rightarrow$ **Fail to reject H_0** $\rightarrow$ unit root $\rightarrow$ **NOT stationary**.

$\ln(X)$: $t = -1.78 > -3.41 \rightarrow$ **Fail to reject** $\rightarrow$ unit root $\rightarrow$ **NOT stationary**.

(b) $\Delta \ln(Y)$: $t = -8.73 < -2.86$ (V1, 5%) $\rightarrow$ **Reject H_0** $\rightarrow$ stationary ✓.

$\Delta \ln(X)$: $t = -9.12 < -2.86 \rightarrow$ **Reject H_0** $\rightarrow$ stationary ✓.

(c) After differencing, the trend is removed. Growth rates ($\Delta \ln$) fluctuate around a roughly constant mean with no visible trend. Using V2 unnecessarily would cost a degree of freedom.

(d) DL(2): $\beta_1 = 0.88$, not significant $\rightarrow$ drop lag 2. DL(1): $\beta_1 = 2.51$, significant $\rightarrow$ stop. Optimal $q = 1$.

(e) $\beta_{11} = 0.28$: A 1pp rise in investment growth this quarter $\rightarrow$ **+0.28pp GDP growth this quarter**, holding last quarter's investment growth constant.

$\beta_{12} = 0.19$: A 1pp rise in investment growth last quarter $\rightarrow$ **+0.19pp GDP growth this quarter**, holding this quarter's constant.

(f) Total multiplier = $0.28 + 0.19 = 0.47$. A permanent 1pp increase in government investment growth is associated with a **0.47pp increase in long-run GDP growth**.

(g) LMSC test:

$H_0: \rho = 0$ (no autocorrelation) vs $H_1: \rho \neq 0$ (autocorrelation present)

Auxiliary regression: $e_t = \alpha_0 + \alpha_1 \Delta \ln(X_t) + \alpha_2 \Delta \ln(X_{t-1}) + \alpha_3 e_{t-1} + u_t$

$LM = 101 \times 0.018 = 1.818$

$1.818 < 3.841 \rightarrow$ **Fail to reject H_0** .

Conclusion: No significant autocorrelation detected at 5%. A3 is plausibly satisfied. The DL(1) model on first-differenced log variables is well-specified. ✓

ONE-PAGE SUMMARY

Stage	Key Actions
FOUNDATIONS	Regression: $Y = \alpha + \beta X + \varepsilon$. OLS finds β . R^2 measures fit (0–1). $t = \beta/SE$ tests if $\beta \neq 0$. $p < 0.05 \rightarrow$ significant. CI: $\beta \pm 1.96 \times SE$. CI excludes zero $\leftrightarrow$ significant at 5%. Five classical assumptions must hold.
RECOGNISE	Time series: order matters. Define lags X_{t-k} and differences ΔX_t . $n > 100$ ideal (min ~35). Use real (deflated) monetary values.
STABILISE	Time plot + Dickey-Fuller test on every variable. Deterministic trend ($\lambda \neq 0$, $ \rho < 1$)? $\rightarrow$ Add t as regressor. Stochastic trend / unit root ($ \rho = 1$)? $\rightarrow$ Take first differences. DF uses special critical values: -2.86 (V1) or -3.41 (V2). NOT ± 1.96 .
MODEL	DL(q): top-down selection. Total multiplier = $\Sigma \beta$. AR(1): total multiplier = $\beta/(1-\gamma)$. REQUIRES $ \gamma < 1$. Interpret: temporary vs. maintained change; levels vs. log-returns.
DIAGNOSE	LMSC: auxiliary regression + lagged residual. LM = $n \cdot R^2 \sim \chi^2(1)$. Reject $H_0 \rightarrow$ autocorrelation $\rightarrow$ re-specify (prefer DL over AR). DW: quick check only — not valid for AR models.